Introduction to Epidemiology

APHS 20203 • Epidemiology & Biostatistics

Course Learning objectives

1. Define epidemiology and key associated terminology, especially confounding.
2. Distinguish among association, prediction, and causation.
3. Explain the Rothman model of sufficient-component causes (RMSCC) and the concept of multifactorial causation.
4. Explain the limitations of anecdotal evidence in drawing epidemiologic conclusions.
5. Explain why disseminating risk information is sometimes necessary but rarely sufficient to improve population health.

Lecture Learning objectives

By the end of this lesson, you'll be able to:

1. Define epidemiology and explain its core assumptions.
2. Describe key events in the history and development of epidemiology.
3. Differentiate epidemic, endemic, and pandemic patterns of disease occurrence.
4. Describe at least three applications of epidemiology in public health practice.

Opening Question

What is Epidemiology?

Respond here

What is Epidemiology?

EPIDEMIOLOGY

- epi – “on, upon”
- demo – “people”
- ology – “study of”

Definition: Epidemiology is concerned with the distribution and determinants of health and diseases, morbidity, injuries, disability, and mortality in populations. Epidemiologic studies are applied to the control of health problems in populations (Quinlan 2025).

Scenario Exercise

1. **Scenario:** A city health department notices that one neighborhood has twice the rate of asthma ER visits in another neighborhood in the same city. This difference has been observed every year for the past three years.
2. **Questions:** Do you think this difference is mostly due to random chance—yes or no? If it’s not random, what might explain the difference?
3. **What are we assuming if we believe those factors matter?**

Assumptions of Epidemiology

ASSUMPTION 1

Human disease does not occur at random

Disease frequency and distribution vary because of identifiable factors that increase or decrease risk.

ASSUMPTION 2

Determinants of disease can be identified

Causal and preventive factors can be studied through systematic investigation of populations.

Descriptive and Analytic Epidemiology

DISTRIBUTION

Descriptive epidemiology

- When was the population affected?
- Where was the population affected?
- Who was affected?

DETERMINANTS

Analytic epidemiology

- How was the population affected?
- Why was the population affected?

Micro-Activity: Thinking Like an Epidemiologist

1. Watch the short video on John Snow and cholera. John Snow: Pioneer of Epidemiology
2. As you watch, focus on these questions:
   What patterns did Snow observe? *(Who, where, when — descriptive epidemiology)*
3. What explanation was he testing? *(How, why — analytic epidemiology)*
   Descriptive epidemiology often comes first; analytic epidemiology builds on it.

History of Epidemiology: Key Events

Event	Time Period	Description
Bubonic Plague Epidemics (Black Death)	1347 - 1351	One of the deadliest pandemics, causing 75–200 million deaths in Eurasia.
Development of Toxicology	16th Century	Paracelsus, “Father of Toxicology,” introduced the principle: “The dose makes the poison.”
Development of Biostatistics	18th Century	John Graunt pioneered statistical analysis, using Bills of Mortality to study disease patterns in London.
Development of Smallpox Vaccine	1796	Edward Jenner created the first vaccine by using cowpox to confer immunity to smallpox.
1918 Influenza Pandemic	1918-19	The “Spanish flu” infected a third of the global population, killing 50–100 million.
Identification of Smoking as a Cause of Cancer	1950	Landmark studies linked smoking to lung cancer, driving public health campaigns against tobacco use.
Eradication of Smallpox	1980	WHO declared smallpox eradicated, marking the success of global vaccination efforts.

Disease Distribution Patterns

01

Endemic

Disease or condition consistently present in a population

- Relatively stable over time
- Expected level of occurrence in a given place

02

Epidemic (Outbreak)

Disease occurrence in excess of what is normally expected

- Sudden increase over a short period
- Often limited to a specific region

03

Pandemic

An epidemic that spreads across multiple countries or continents

- Global distribution
- Affects a large proportion of the population

Exercise: Disease Distribution Patterns

Ebola virus cases in parts of Africa exceed what is normally expected for that region. In this situation, the Ebola virus was [a/an] __________.

A. endemic

B. epidemic

C. pandemic

Respond here

Exercise: Disease Distribution Patterns

Malaria is always present in Africa due to the presence of infected mosquitoes. In this situation, malaria is [a/an] _________.

A. endemic

B. epidemic

C. pandemic

Respond here

Exercise: Disease Distribution Patterns

Between 2019 and 2023, the SARS-CoV-2 virus infected more than 600 million people worldwide. In this situation, the SARS-CoV-2 virus was [a/an] __________.

A. endemic

B. epidemic

C. pandemic

Respond here

Use of Epidemiology

In the first group presentation for this course, you will:

Describe a health problem

(Who is affected? Where? When?)

Explain possible reasons

(Risk factors or exposures)

Interpret evidence

(Patterns and associations)

Propose prevention or control strategies

References

Quinlan, Scott. 2025. *Friis' Epidemiology 101*. 3rd ed. Jones & Bartlett Learning.